(a) 4 (b) 16
(c) 25 (d) 36

Ans. (d) : In every figure the sum of square route of corner digit comes in middle box.
Here, underfoot of 9 = 3
And, underfoot of 64 = 8, underfoot of 1 = 1
Now, 3 + 8 + 1 = 12
Then, 18 – 12 = 6
So, $X = (6)^2$
= 36 Ans.

119. Consider the following statements:

1. According to the Census 2001, Kerala has the smallest gap in male and female literacy rates among the 28 states of India (Delhi and Pondicherry not included).

2. According to the Census 2001, Rajasthan has literacy rate above the national average literacy rate.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Mizoram is the state where male literacy is 90.7% and female literacy is 86.7%. So, Mizoram is the state of smallest gap in male and female literacy, where Kerala has 94.2% in male and 87.7% in female. In statement 2 literacy of Rajasthan is 60.4%, where national literacy is 64.8%.

120. Zayed bin Sultan-al-Nahyan was a long serving President of which of the following?

(a) Oman
(b) Kuwait
(c) United Arab Emirates
(d) Saudi Arabia

Ans. (c) : Zayed bin Sultan Al Nahyan was the principal driving force behind the formation of the United Arab Emirates (UAE), the ruler of Abu Dhabi and first President of the United Arab Emirates, a post which he held for over 33 years (1971-2004).

121. In which State is the Rajiv Gandhi National Institute of Youth Development located?

(a) Tamil Nadu (b) Karnataka
(c) Himachal Pradesh (d) Uttaranchal

Ans. (a) : The Rajiv Gandhi National Institute of Youth Development, Sriperumbudur, Tamil Nadu, is an Institution of National Importance by the Act of Parliament No. 35/2012 under the Ministry of Youth Affairs & Sports, Government of India. The Rajiv Gandhi National Institute of Youth Development was set up in 1993 under the Societies Registration Act, XXVII of 1975.

122. Who among the following is a Hindustani classical singer?

(a) Geeta Chandran (b) Leela Samson
(c) Gangubai Hangal (d) Swapnasundari

Ans. (c) : Gangubai Hangal was a Hindustani classical singer.

123. Match List-I (Person) with List-II (Area of Distinction) and select the correct answer using the code given below the Lists:

List-I (Person)	List-II (Area of Distinction)
A. E.C.G. Sudarshan	1. Theatre
B. V. Shanta	2. Violin
C. T.N. Krishanan	3. Research in Physics
D. Mahesh Dattani	4. Cancer treatment
	5. Water harvesting

Codes:

	A	B	C	D
(a)	3	1	2	4
(b)	2	4	5	1
(c)	3	4	2	1
(d)	2	1	5	4

Ans. (c) : A-3; B-4; C-2; D-1

A. E.C.G. Sudarshan	Research in Physics
B. V. Shanta	Cancer treatment
C. T.N. Krishanan	Violin
D. Mahesh Dattani	Theatre

124. What is the Universal Product Code (UPC) adopted for?

(a) Fire safety code in buildings
(b) Earthquake-resistant building code
(c) Bar code
(d) Against adulteration in eatables

Ans. (c) : The Universal Product Code UPC is a barcode symbology i.e. a specific type of barcode that is widely used in the United States, Canada, United Kingdom, Australia, New Zealand and in other countries for tracking trade items in stores.

125. Consider the following statements:

1. The Richter scale is a logarithmic scale and so an increase of 1 magnitude unit represents a factor of 10 times in amplitude.

2. Each integer reading of the Richter scale has an energy 100 times that of the previous integer reading.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : The magnitude of intensity of energy released by an earthquake is measured by Richter Scale. The number indicating magnitude on Richter Scale ranges between 0 and 9 but infact the scale has no upper limited of number because it is a logarithmic scale. But each integer reading of the Richter Scale has in energy 32 times of previous integer reading. So statement 2 is not correct.

126. For which sport is the Val Barker Cup awarded in the Olympic Games?

(a) Swimming (b) Boxing
(c) Long Jump (d) High Jump

Ans. (b) : Val Barker Cup is awarded for boxing in the Olympic Games.

127. How many numbers are there in all from 6000 to 6999 (Both 6000 and 6999 included) having at least one of their digits repeated?

(a) 216 (b) 356
(c) 496 (d) 504

Ans. (c) : If we subtract the number where no digits are repeating from total number between 6000 and 6999 we get the answer.
Total No = 1000
6 is fixed in the 1st place and the second place can have 9 different possibilities, then 3rd place can have 8 and 4th place 7. This will result in numbers where no digit is repeating i.e.
$1 \times 9 \times 8 \times 7 = 507$
Subtracting from 1000 we get 496.
Hence there will be 496 such possibilities.

128. Each of 2 women and 3 men is to occupy one chair out of 8 chairs, each of which is numbered from 1 to 8. First, women are to occupy and two chairs from those numbered 1 to 4; and then the 3 men would occupy any three chairs out of the remaining 6 chairs. What is the maximum number of different ways in which this can be done?

(a) 40 (b) 132
(c) 1440 (d) 3660

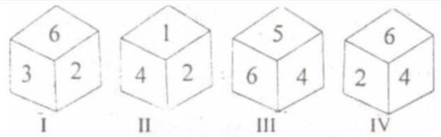
Ans. (c) : Two women can occupy 2 chairs out of the first four chairs in ${}^4P_2$ ways.
3 men can occupy the remaining 6 chairs in ${}^6P_3$ ways
$\therefore$ total no. of ways $= {}^4P_2 \times {}^6P_3 = 1440$

129. "Lectures from Colombo to Almora" is based on the experiences of which one of the following?

(a) Veer Savarkar
(b) Annie Besant
(c) Ramkrishna Paramhansa
(d) Swami Vivekanand

Ans. (d) : Lectures from Colombo to Almora is a book of Swami Vivekananda based on the lectures he delivered in Srilanka and India after his return from the West.

130.



Each of the six faces of a cube is numbered by one of the six digits from 1 to 6. This cube is shown in its four different positions in the figures I, II, III and IV. Consider the following statements:

1. Figures II and III are sufficient to know as to which face is opposite to the face numbered 6.

2. Figures II and III are sufficient to know as to which face is opposite to the face numbered 4.

3. Figures I and IV are sufficient to know as to which face is opposite to the face numbered.

Which of the statements given above are correct

(a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3, only (d) 1, 2 and 3

Ans. (d) : 1 is correct as it clearly shows that 1 and 5 are the top and bottom 2 and 6 are on the sides 4 so 2 is opposite 6.



- 2 is correct on the same lines, 3 is opposite 4.
- 3 is correct as I & IV clearly tells that 4 is apposite 3. All the three statements regarding the figure given in the question are correct. Hence, the answer is (d)

131. Which one of the following pairs is not correctly matched?

(a) Sigmund Freud : Psychoanalysis
(b) Anna Freud : Child psychiatry
(c) Milton Friedman : Economics
(d) Eric R. Kandel : Literature

Ans. (d) : Eric Richard Kandel is an American neuropsychiatrist. He was a recipient of the 2000 Nobel Prize in Physiology or Medicine for his research on the physiological basis of memory storage in neurons.

132. Projects and Development India Limited is mainly engaged in design engineering, procurement and supervision of construction/commissioning in which area?

(a) Pharmaceutical plants
(b) Petroleum refineries and sugar factories
(c) Ship-building yards
(d) Fertilizer and allied chemical plants